

Homework 1

Q1. If $f(x) = b^x$, $b > 0$ and $b \neq 1$ then _____ is the asymptote. [2]

(a) $y = 0$

(b) $y = 1$

(c) $y = x$

(d) $x = 0$

Q2. Which of the following is not a polynomial function? [2]

(a) $f(x) = 3x^2 + 2x + 1$

(b) $f(x) = \frac{1}{x} + 4$

(c) $f(x) = x^3 - 5x$

(d) $f(x) = 7x^4 + 6$

Q3. What is the effect of the transformation $y = 3f(x)$ on the graph of $f(x)$? [2]

(a) Horizontal stretch

(b) Vertical stretch

(c) Horizontal compression

(d) Vertical compression

Q4. In a distance-time graph, a horizontal line indicates: [1]

(a) Constant speed

(b) Constant acceleration

(c) Object at rest

(d) Increasing speed

Q5. If $f(x) = |x + 3|$, the vertex of the graph is: [2]

(a) $(-3, 0)$

(b) $(3, 0)$

(c) $(0, 3)$

(d) $(0, -3)$

Q6. Find the natural numbers a and b such that $b > a$ and [2]

$$a^b = b^a$$

(a) $a = 4, b = 2$

(b) $a = 2, b = 4$

(c) $a = 2, b = 2$

(d) None of the above

Q7. Find the value of x that solves the equation [2]

$$4 \ln(x) = \ln(8x)$$

(a) $x = -2$

(b) $x = e^{-2}$

(c) $x = 2$

(d) None of the above

Q8. Which choice is the solution to the inequality: $x^2 + 2x - 15 < 0$? [2]

(a) $-3 < x < 5$

(b) $-5 < x < 3$

(c) $x < -5$ or $x > 3$

(d) $x < -3$ or $x > 5$

Q9. Solve the quadratic equation $x^2 - 5x + 6 = 0$ for x . [2]

(a) $x = 1$ and $x = 6$

(b) $x = -1$ and $x = -6$

(c) $x = 2$ and $x = 3$

(d) $x = -2$ and $x = -3$

Q10. For the function $f(x) = \sin x$, show using trigonometric identities that:

[3]

$$\frac{f(x+h) - f(x)}{h} = \sin(x) \left[\frac{\cos(h) - 1}{h} \right] + \cos(x) \left[\frac{\sin(h)}{h} \right]$$

Q10:- prove $\frac{f(x+h) - f(x)}{h} = \sin(x) \left[\frac{\cos(h) - 1}{h} \right] + \cos(x) \left[\frac{\sin(h)}{h} \right]$

when $f(x) = \sin(x)$.

$$\frac{f(x+h) - f(x)}{h} = \frac{\sin(x+h) - \sin(x)}{h}$$

$$\because \sin(u+v) = \sin u \cos v + \cos u \sin v$$

$$\Rightarrow \frac{\sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x)}{h}$$

$$\Rightarrow \frac{\sin(x) \cos(h) - \sin(x) + \cos(x) \sin(h)}{h}$$

$$\Rightarrow \frac{\sin(x) \cos(h) - \sin(x)}{h} + \frac{\cos(x) \sin(h)}{h}$$

$$\frac{f(x+h) - f(x)}{h} \Rightarrow \sin(x) \left[\frac{\cos(h) - 1}{h} \right] + \cos(x) \left[\frac{\sin(h)}{h} \right]$$

Hence proved!